

Sepehr Rezaee

sepehrrezaee2002@gmail.com | github.com/SepehrRezaee | My Scholar | linkedin.com/in/sepehr-rezaee/
sepehrrezaee.github.io

Education

Shahid Beheshti University, BS in Computer Sciences	2021 – 2025
• Interests: Deep Learning, Computer Vision, Generative AI, and AI Safety	
Allameh Tabatabaei (Advanced) High School, Math Diploma	2019 – 2021
• GPA: 3.87/4.0	

Experience

AI Specializing In Applied LLMs PropTy Global	2024 – Present
- Led the development of an AI-powered chatbot and home finder system using open-source LLMs and Retrieval-Augmented Generation (RAG) to enhance real estate customer service. - Engineered a scalable backend infrastructure with FastAPI, PostgreSQL, and MongoDB, ensuring high reliability and performance for real-time interactions. - Integrated and fine-tuned LLMs (GPT-Neo, GPT-J, LLaMA) for domain-specific tasks, significantly improving the accuracy and contextual relevance of chatbot responses. - Successfully deployed the solution with Docker and Kubernetes, achieving seamless scalability and handling a large volume of concurrent user requests. - Implemented real-time monitoring and continuous improvement strategies using Prometheus, Grafana, and the ELK stack, ensuring system stability and performance.	
Research Assistant, Robust and Interpretable Machine Learning Lab – Sharif University of Technology, Tehran	2024 – 2025
- Authored and co-authored 3 papers submitted to NeurIPS 2024, focusing on enhancing model reliability and security in machine learning. - Developed and implemented 3 robust machine learning pipelines, increasing model reliability under adversarial conditions. - Collaborated with a multidisciplinary team of 10 members to integrate machine learning solutions into real-world applications (Autonomous Driving, Face Detection, Diagnosing Disease), improving operational efficiency. - Presented research findings at 2 international conferences, elevating the lab's visibility and fostering academic collaborations.	
Research Assistant, Artificial Intelligence and Scientific Computing Lab – Shahid Beheshti University, Tehran	2023 – 2025
- Co-authored 2 under-review & 1 published research papers, including: - Physics-Informed Lane-Emden Solvers Using Lynx-Net: Implementing Radial Basis Functions in Kolmogorov Representation - Leveraging Physics-Informed Convolutional Neural Networks (PICNNs) to Solve Linear and Non-linear Fokker-Planck Equations (FPEs) - Comparison of Pre-training and Classification Models for Early Detection of Alzheimer's Disease Using Magnetic Resonance Imaging - Modeled disease progression using differential equations, enhancing the understanding of biological mechanisms. - Employed Physics-Informed Neural Networks (PINNs), increasing model accuracy through the integration of physical laws.	
Deep Learning and Neuroscience Intern Researcher, Institute for Research in Fundamental Sciences (IPM) – Tehran	2023 – 2024
- Conducted comprehensive M/EEG data analysis utilizing advanced deep learning techniques to decode neural signals. - Developed and optimized neural network architectures for improved signal processing and feature extraction. - Collaborated with neuroscientists to interpret data results and contribute to the understanding of brain functionalities. - Assisted in the preparation of research manuscripts and presentations for academic dissemination.	

Publications

• Scanning Trojaned Models Using Out-of-Distribution Samples (Accepted to NeurIPS)	2024
Authors: Hossein Mirzaei, Ali Ansari*, Bahar Dibaei Nia*, Mojtaba Nafez†, Moein Madadi†, Sepehr Rezaee †, Zeinab Sadat Taghavi, Arad Maleki, Kian Shamsaie, Mahdi Hajjalilue, Jafar Habibi, Mohammad Sabokrou, Mohammad Hossein Rohban	
• A Contrastive Teacher-Student Framework for Novelty Detection Under Style Shifts (Submitted to ICLR)	2025
Authors: Hossein Mirzaei, Mojtaba Nafez, Moein Madadi, Arad Maleki, Mahdi Hajjalilue, Zeinab Sadat Taghavi, Sepehr Rezaee , Ali Ansari,	

- **Comparison of Pre-Training and Classification Models for Early Detection of Alzheimer's Disease Using Magnetic Resonance Imaging (Accepted in ICCCCC 2023)** 2023
Authors: AH Karami, S Rezaee, E Mirzabeigi, K Parand
- **Hierarchical Clustering Algorithms, Chapter of Unsupervised Algorithms: Clustering (with Implementation)** Aarvan Publications 2022
Authors: Kourosh Parand, Sepehr Rezaee, et al.

Selected Projects

- AI Model Security: Enhancing Robustness Against Backdoors and Trojans** 2024
 - Developed methods to detect and mitigate backdoors in machine learning models, enhancing AI deployment security.
 - Engineered algorithms using statistical analysis and pattern recognition, improving trojan detection rates.
 - Contributed to NeurIPS 2024 publications, advancing the field of AI model security.
 - **Tools Used:** Python, PyTorch, Scikit-learn, LaTeX
- AI-Based Application for Early Detection of Alzheimer's Disease** 2023 – 2024
 - Designed and implemented a customized multi-modal model integrating biomedical and MRI datasets.
 - Enhanced diagnostic accuracy through advanced machine learning techniques with Vision Language Models (VLMs).
 - **Tools Used:** PyTorch, Hugging Face, OpenCV
- Physics-Informed Neural Networks for Disease Progression Modeling** 2023
 - Created a Physics-Informed Neural Network integrating differential equations to predict disease progression accurately.
 - Utilized clinical datasets and validated models with patient data, achieving higher accuracy than traditional methods.
 - Published findings in peer-reviewed journals, contributing to AI-based healthcare innovations.
 - **Tools Used:** PyTorch, NumPy, SciPy, Pandas
- AI-Driven M/EEG Data Analysis for Neuroscience Research** 2022
 - Applied deep learning techniques to decode M/EEG signals, uncovering neural mechanisms.
 - Streamlined data workflows by automating preprocessing and artifact removal, enhancing analysis efficiency.
 - Facilitated insights into brain connectivity, supporting high-impact neuroscience research publications.
 - **Tools Used:** MNE-Python, PyTorch, NumPy, Pandas

Awards & Honors

- Winner of the Best Ideator Award (The 7th National Young Scientists Festival)** 2023
For designing an AI-based assistant for the early detection of Alzheimer's disease.
- Placed 352nd out of approximately 150,000 students in the national entrance exam** 2020

Teaching Assistant

- Advanced Programming Head Teaching Assistant**, Shahid Beheshti University, Tehran 2024 – Present
- Data Mining and Analysis Head Teaching Assistant**, Shahid Beheshti University, Tehran 2023
- Basic Programming Teaching Assistant**, Shahid Beheshti University, Tehran 2022
- Assistant Teacher and Mentor** 2022 – 2023
 - Applications of Data Science and Artificial Intelligence in the Petrochemical Industry, the Water Industry & the Electricity Industry

Selected Courses

Courses: Foundations of Data Science (A^+ , 1st), Data Mining (A^+ , 1st), Advanced Data Mining (A^+ , 1st), Foundation of Numerical Analysis (A^+ , 1st), Non-Linear Optimization (A^+ , 1st), Partial Differential Equations (A^+ , 1st), Electromagnetics (A^+ , 1st), Neural Network (A^+ , 3rd), Foundation of Logic and Set Theory (A^+ , 3rd), Principles of Operating Systems (A^+ , 2nd), Foundations of Machine Learning (A^+ , 2nd), Elements of Probability (A , 4th), Data Structures & Algorithms (A , 5th)

Skills

Programming Languages: Python, C++, C, MATLAB, C# & Java

Python Frameworks & Libraries: PyTorch, TensorFlow, OpenCV, MNE-Python, NumPy, SciPy, Matplotlib, Scikit-Learn, NiPype, FastAPI, Django, Django REST Framework, Selenium

Other Tools and Technologies: JAX, PostgreSQL, NoSQL, MongoDB, Kotlin, , Git, Docker, Linux, Bootstrap

Interpersonal Skills: Problem Solving, Team Working

Languages: Fluent in Persian (speaking, reading, and writing), English (Professional working proficiency)

Reference Contacts

Prof. Kourosh Parand - k_parand@sbu.ac.ir

Prof. Mohammad Hossein Rohban - rohban@sharif.edu

Prof. Mohammad Sabokrou - mohammad.sabokrou@oist.jp